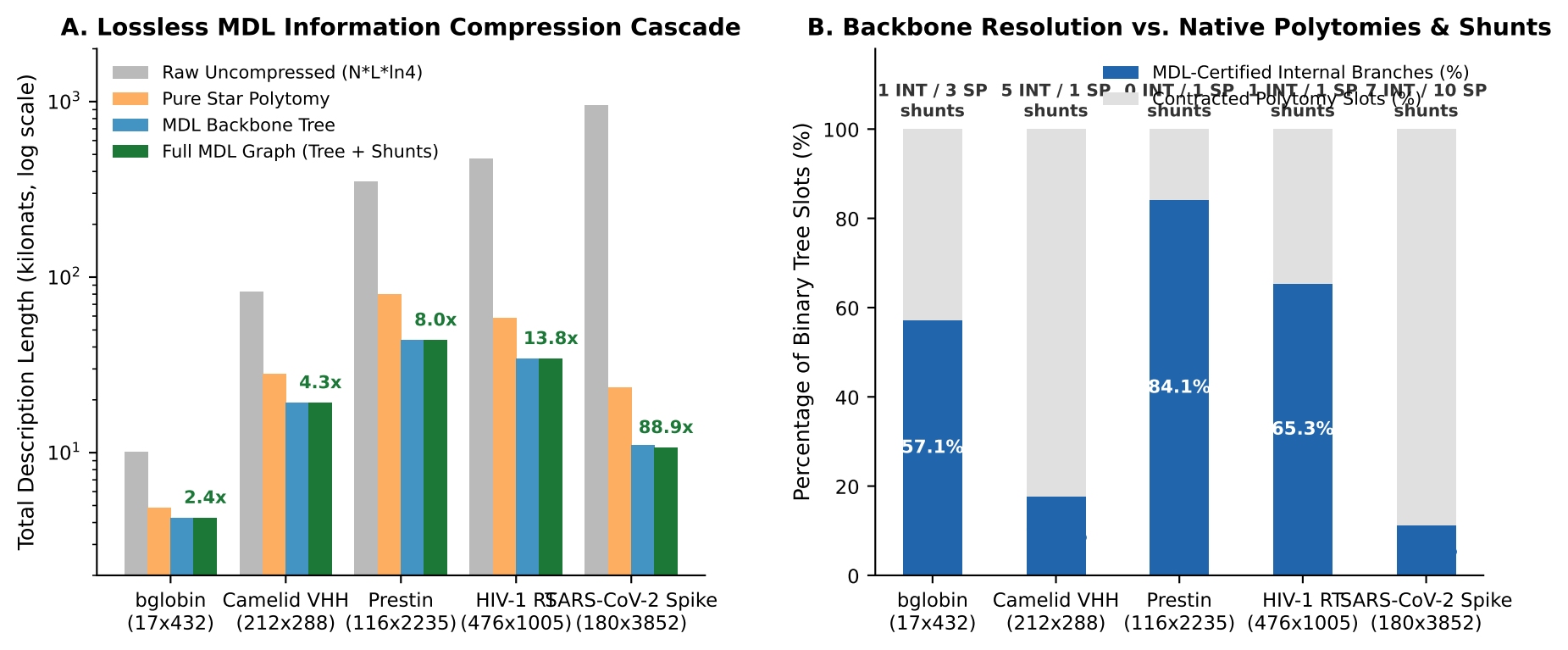
A. Lossless MDL Information Compression Cascade
Raw Uncompressed (N*L*ln4)
Pure Star Polytomy
MDL Backbone Tree
Full MDL Graph (Tree + Shunts)
Total Description Length (kilonats, log scale)
10^3
10^2
10^1
2.4x
4.3x
8.0x
13.8x
88.9x
bglobin (17x432)
Camelid VHH (212x288)
Prestin (116x2235)
HIV-1 RT (476x1005)
SARS-CoV-2 Spike (180x3852)
B. Backbone Resolution vs. Native Polytomies & Shunts
MDL-Certified Internal Branches (%)
Contracted Polytomy Slots (%)
Percentage of Binary Tree Slots (%)
1 INT / 3 SP shunts
5 INT / 1 SP shunts
0 INT / 1 SP shunts
1 INT / 1 SP shunts
7 INT / 10 SP shunts
57.1%
84.1%
65.3%
0
20
40
60
80
100
bglobin (17x432)
Camelid VHH (212x288)
Prestin (116x2235)
HIV-1 RT (476x1005)
SARS-CoV-2 Spike (180x3852)